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MILITARY SPECIFICATION

INDICATOR, AZIMUTH-ELEVATION-RANGE, IP-446/APG-53A

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be a radar indicator suitable for installation and use in high-performance carrier-based or land-based aircraft. When properly installed as a component of Radar Set AN/APG-53A, the indicator shall display range, range versus elevation angle, or range versus azimuth angle in accordance with signals derived from other components of Radar Set AN/APG-53A.

1.2 Classification - The Indicator, Azimuth-Elevation-Range, IP-446/APG-53A shall be of one type and shall consist of the following:

Item	Type Designation	Max. Allow. Size (Inches) W	D	L	Max. Allow. Wt. (Lbs)	Appl. Para.
Indicator, Azimuth-Elevation-Range	IP-446/APG-53A	5.750 (See figure 3)	5.750	16	14	3.3

1.3 Auxiliary Equipment - The indicator has been designed to be used as a part of Radar Set AN/APG-53A, and to derive signals and low-voltage power from other parts of the radar set.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

Military

MIL-P-80	Plastic Sheet, Acrylic Base, Antielectrostatic, Transparent (For Indicating Instrument Windows)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-P-5425	Plastic; Sheet Acrylic, Heat Resistant
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting

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MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-L-006805	Lacquer, Camouflage
MIL-E-7894	Electric Power, Aircraft Characteristics of
MIL-L-25467	Lighting, Integral, Instrument, General Specification For
MIL-P-17555	Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection, For Aircraft Electronic Equipment, Format For
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

STANDARDS

Military

MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, or 115/200 Volt 3-Phase AC, or Combinations
MS 28053	Meter, Time Totalizing, 400-cycle

PUBLICATIONS

Bureau of Aeronautics

SAR-300	Material Changes and Material Bulletins, Preparation of

2.2 Availability of Documents:

(1) When requesting applicable documents, give both title and number of each. Requests for all documents required in connection with specific procurement functions shall be made via the cognizant Inspector of Naval Material, Bureau of Aeronautics Representative, or Contracting Officer.

(a) Copies of this specification, applicable specifications, and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

(b) Copies of applicable drawings may be obtained upon application to the Commanding Officer, U. S. Naval Avionics Facility, Indianapolis 18, Indiana.

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2.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts, Materials and Processes - In the selection of parts, materials and processes, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts, materials and processes as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specification.

(3) When a nonstandard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event that the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400, except as follows:

(1) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of Specification MIL-E-5400 are not met.

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(2) Nonstandard parts of an electrical nature as defined in Specification MIL-E-5400, shall be approved before the preproduction model of the equipment is submitted for tests. The request for approval shall be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Nonstandard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

(3) Tubes (Electron) - Electron tubes shall not be used except where the use of semi-conductors would not be practicable.

(4) Dissimilar Metals - In certain applications, such as in lubricated gears and ball bearings requiring intimate contact between dissimilar metals, and where passivation is impractical, dissimilar metals may be used in intimate contact.

(5) Plastic Window - Pending qualified suppliers of plastic windows in accordance with Specification MIL-P-80, material in accordance with Specification MIL-P-5425 may be used.

(6) Ferrous Alloys - Passivation shall not be required for enclosed lubricated mechanical working parts made of corrosion resistant ferrous alloys where close tolerances must be held, such as gears, bearings, cams and shafts.

3.1.2 Electron Devices - The choice and application of electron devices used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any electron device not meeting the requirements of (1) above must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor shall submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.2.1 Electron Device Complement - An electron device complement report shall be prepared in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 Class 2 equipment for design, construction, and workmanship, unless otherwise specified herein.

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3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 14 pounds.

3.2.3 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.4 Service Conditions - The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as follows:

3.2.4.1 Vibration - When mounted directly to a mounting panel without vibration isolators, the equipment shall operate within specified performance limits when the mounting panel is submitted to continuous sinusoidal vibration within the frequency range of 5 to 500 cycles per second and having a double amplitude as defined by figure 1.

3.2.4.2. Gunfire - When mounted directly to a mounting panel, the equipment shall operate within specified performance limits while the mounting plate is subjected to 6000 impact shocks representative of gunfire, consisting of 1000 shocks of 25g ± 10% peak magnitude in opposite directions along each of 3 mutually perpendicular axes, each shock impulse having a time duration of 5 milliseconds ± 50%.

3.2.4.3 Crash Safety - When mounted directly to a mounting panel, the equipment shall withstand at least 12 impact shocks of 30g ± 10% peak magnitude applied to the mounting plate, consisting of 2 shocks in opposite directions along each of 3 mutually perpendicular axes and each shock impulse having a time duration of 11 milliseconds ± 10%. Bending and distortion shall be permitted; however, there shall be no failure of the attaching points and the equipment shall remain in place.

3.2.4.4 Design Load Factor - When mounted directly to a mounting panel, the equipment shall be capable of withstanding a continuous load factor of 7.0, caused by aircraft accelerations, in opposite directions along each of three mutually perpendicular axes. Permanent degradation in performance or structural yielding shall not occur below 115 per cent of the design load factor. Structural failure shall not occur below 150 per cent of the design load factor.

3.2.4.5 Altitude - The altitude range shall be 0 - 60,000 feet (30 inches of mercury to 2 inches of mercury).

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3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400. Unless otherwise specified, interchangeability shall be measured against a model supplied by the Bureau of Aeronautics.

3.2.6 Reliability - The equipment shall operate within the environmental and performance limits of this specification continuously or intermittently for at least 200 hours without internal maintenance adjustments, or replacement of parts. The equipment shall provide a total life of 2000 hours, with reasonable maintenance adjustments and replacement of parts.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7.1 Tools - Special tools, as required, shall be supplied in a separate kit as part of this equipment, and shall be kept to an absolute minimum. A list of special tools shall be submitted to the procuring agency.

3.2.8 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.9 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the Bureau of Aeronautics prior to preproduction testing.

3.2.10 Cabling and Connections -

3.2.10.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.10.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.11 Interference - The interference requirements of Specification MIL-I-6181 shall apply with the use of a minimum of interconnecting shielded conductors.

3.2.12 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3 Detail Requirements

3.3.1 Indicator, Azimuth-Elevation-Range IP-446/APG-53A - The indicator shall meet the following requirements and shall give satisfactory performance when installed as a component of the Radar Set AN/APG-53A.

3.3.1.1 Display - The indicator shall provide circuitry as required for the display modes defined in the succeeding paragraphs. Range and mode switching shall be accomplished by providing six external connections, which may be grounded by external switching for various combinations of range and mode.

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3.3.1.1.1 Range Only - The range only mode shall display a horizontal line approximately half the width of the display aperture centered laterally about the vertical centerline of the display aperture. This horizontal line shall be deflected vertically to indicate range from 0 to 10 nautical miles. The line width shall be 1/16-inch ± 1/32-inch and the lower edge shall be used for range measurements. The line shall be straight and horizontal within 1/32-inch throughout the range of vertical deflection. The input signal for the range-only mode will be a d-c voltage having a scale factor of 20 volts per nautical mile. The load impedance imposed on the range voltage source shall be in excess of one megohm.

3.3.1.1.2 Elevation-Range - The elevation-range mode shall display elevation angles from 10 degrees above the airplane velocity vector to 15 degrees below the velocity vector against linear horizontal slant range scales of 0 to 10 and 0 to 20 nautical miles. Range increases from left to right. The 0 degree elevation reference line shall be horizontal and located vertically as shown in figure 2. The display shall be intensity modulated.

(1) Input signals for the elevation-range mode will be as follows:

Trigger - 93 ohms impedance; plus 50 ± 15 volts; 0.6 to 1.0 microsecond duration

Video - 100 ohms impedance; the input will have a dynamic range of 20 db varying from .2 V to 2 volts. In the elevation - range mode an altitude clearance marker will be mixed with the video. The clearance marker, which varies from 1 microsecond width and .6 volt amplitude at 1000 feet csc 15° delay to 20 microsecond width and 2.0 V amplitude at full 20 mile range, shall write to full brightness in one vertical sweep cycle.

Antenna position - Elevation angle; two (push-pull) 50,000-ohm, 2-watt, linear potentiometers. Excitation for these potentiometers will be furnished by the auxiliary power supply defined in paragraph 3.3.1.5 at a voltage level of +130 V DC as shown in Figure 4.

3.3.1.1.3 Range Azimuth - Range versus azimuth shall be displayed in two externally-selected forms as defined below. Each of these shall display slant range (nautical miles) on linear vertical scales and azimuth angle on linear horizontal scales. The displays shall be intensity modulated in accordance with the level of video input signals.

3.3.1.1.3.1 0 to 10, 0 to 20 Mile Horizontal Terrain Clearance - This mode shall display slant range versus azimuth angles of ± 30 degrees. Input signals shall be as follows:

Trigger - 93 ohms impedance, plus 50 ± 15 volt amplitude, 0.6 to 1.0 microsecond duration

Video - 100 ohms impedance. The dynamic range of the video signals will be 20 db; varying from .2 V amplitude to 2 volts amplitude. The entire dynamic range of these signals shall be displayed at full brightness in one horizontal sweep cycle.

Antenna - Azimuth angle data: Two 50K, 2-watt linear push-pull potentiometers. Potentiometers have ± 150-degree range for radar beam coverage of ± 30 degrees. Potentiometer excitation will be furnished as noted in paragraph 3.3.1.1.2.

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3.3.1.1.3.2 0 to 20, 0 to 40 Mile Search - This mode shall display slant range for azimuth angles of ± 30 degrees. Signals will be as specified in 3.3.1.1.3.1 with the following exceptions:

(a) Range limits change from 0 to 10 and 0 to 20 to 0 to 20 and 0 to 40 nautical miles.

3.3.1.1.4 Brilliance and Contrast - The indicator shall provide the greatest practicable intensity (in excess of 1,000 foot lamberts) and contrast to permit daylight viewing.

3.3.1.1.5 Video Amplifier - The IP-446 video amplifier shall have a gain sufficient to display a 3 microsecond pulse of 0.2 volt amplitude to full brightness on one vertical sweep cycle of the elevation range mode or one horizontal sweep cycle of the range azimuth modes. The frequency at which the gain shall be 3 decibels below the mid frequency gain shall be no lower than 5 megacycles or greater than 750 cps. The video amplifier shall have limiting characteristic so that a 0.2 volt signal and a 2 volt signal will write to approximately the same brightness. The overload characteristic of the amplifier shall be such that a 0.3 microsecond video signal of 0.2 volt amplitude following a 0.3 microsecond signal of 2 volt amplitude by 3 microseconds will not be obscured.

3.3.1.1.6 Storage - The video information shall be stored for a variable length of time which is controlled by the storage potentiometer. The minimum storage time shall be approximately 0.5 second and the maximum storage time approximately 1.5 seconds.

3.3.1.2 Range Flags - Solenoid (or equivalent) operated flags shall be provided to indicate the orientation of the range scale for the various display modes. These flags shall be located and configured in general accordance with figure 2. The flag symbols shall display surfaces coated with camouflage white lacquer No. 601 per Specification MIL-L-006805 when operated and camouflage black lacquer No. 604 per Specification MIL-L-006805 when not operated. In addition, the vertical range scale flag shall indicate the scale calibration of 10, 20, or 40. The profile range scale flag shall indicate the scale calibration of 10 or 20 as selected.

3.3.1.3 Controls - The receiver gain controls shall be concentrically located in the lower left-hand corner. The gain controls shall consist of two 5000-ohm, one-third watt, linear-taper potentiometers. End resistors of 1000 ohms shall be provided in the ground ends. Operating controls for setting the level of brilliance and storage shall be located concentrically in the upper left corner of the indicator. The operating control for setting the level of panel illumination shall be mounted in the lower right corner of the indicator. The controls shall be identified as to their function as shown in figure 2.

3.3.1.4 Lighting -

3.3.1.4.1 Illumination - The fixed indications, as shown in figure 2, and the range flags shall be illuminated for night operation in accordance with Specification MIL-L-25467.

3.3.1.4.2 Intensity - The fixed indications shall be suitable for daylight viewing with an indicator intensity of 1000 foot-lamberts or less.

3.3.1.5 Power Supplied - The indicator shall operate from an auxiliary low-voltage power supply which will be supplied as part of the Radar Set AN/APG-53A. The following voltages will be provided:

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+250 volts ± 1%
+130 volts ± 3%
25 to 29 volts d-c
5.9 to 6.7 volts (rms) 400 cps isolated
-250 volts ± 1%
-600 volts ± 2%
108 to 121 volts (rms) 400 cps
5.9 to 6.7 volts (rms) 400 cps, one side connected to -600 volts
5 volts (rms) 400 cps, lighting

The indicator shall operate satisfactorily within the transient voltage limits noted in Figure 1 over-voltage Specification MIL-E-7894. The indicator shall not be damaged when subjected to voltages between the normal and the over-voltage limits noted in Figure 1 Specification MIL-E-7894.

3.4 Performance

3.4.1 Accuracy - All sweeps shall be accurate to ± 2 per cent over the temperature range of -20°C to +71°C and ± 5% from -54°C to -20°C.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements.

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the Bureau of Aeronautics, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788. The following additional tests and modifications to tests of Specification MIL-T-5422 shall be included:

4.2.3.1 Service Condition Tests - Service conditions tests shall be modified in accordance with the requirements of 3.2.4.

4.2.3.2 Electrical Tests -

(a) Apply power supply voltages as shown in Figure 4.

(b) Ground terminals 10 and 13 of J301 to place the indicator in 10-mile, E-scan condition.

(c) Apply triangular sweep voltages to terminals 1 and 2 in phase opposition, the voltages varying linearly between 0 and 130 volts with a frequency of one cycle per second.

(d) Apply trigger pulses having a repetition rate of 3000 pps, a pulse duration of 1.0 microsecond, and an amplitude of 50 volts, to terminal A.

(e) Apply pulses to terminal B having a repetition rate of 3000 pps, a pulse duration variable between 0.35 and 20 microseconds, and an amplitude variable between 0.5 and 3.0 volts positive. These pulses shall be synchronized with the trigger pulses and occur approximately 60 microseconds after the corresponding trigger pulse.

(f) With the pulses of (e) adjusted to 0.5 volt amplitude and 0.35 microsecond duration, adjust the indicator for optimum contrast and a storage time of 0.5 second. The brilliance measured at the center of the tones shall be in excess of 1000 foot lamberts.

(g) In addition to the video input outlined in (f), mix a video marker pulse whose position with respect to the aircraft velocity vector line on the display is horizontal and positioned 1000 feet below. In order to simulate the marker pulse supplied by the Altitude Computer CP-403/APG-53A during system operation, the following pulse width and delay parameters are to be used in the display of the marker pulse.

(1) The delay of the pulse is to be varied in accordance with the following:

Pulse delay (feet) = 1000 csc antenna depression angle

(2) The width of the pulse shall vary from one microsecond at -15° depression angle to 20 microseconds as the curve intersects the 20 mile range point. Between these points the width shall vary as the cosecant of the antenna depression angle.

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The line on the display generated by this marked must be displayed at full brightness after one vertical sweep cycle and must be an unbroken line on both the 10 and 20 mile ranges.

(h) Change the pulse duration of the video input signal to 20 microseconds and the amplitude to 3 volts. Remove the ground from terminals 10 and 13 and ground terminal 9 and 12. Transfer the sweep voltages of (c) to terminals 20 and 21. Remove the video marker pulse specified in (g).

(j) Observe the waveform of the signal at the output of the video amplifier. The trailing edge overshoot shall not exceed 5 per cent of the amplitude of the output waveform.

(k) Remove the pulse generator from terminal B and substitute a sine wave generator. With the output voltage of the video amplifier held constant at 15 volts peak-to-peak, measure the input voltage as the frequency is varied between 750 cps and 5 megacycles. The total variation in input amplitude shall not exceed 3 decibels.

(m) Adjust the sweep calibration controls for proper calibration of the 10 - mile range. Apply range marks sequentially or simultaneously at 2, 4, 6, 8, and 10 miles to terminal B of J301. The position of each range mark shall be within 1/16 inch of the corresponding calibration mark. This accuracy shall be maintained with no readjustment of controls at -20°C and +71°C.

(n) With the indicator adjusted as in (m), the range marks shall be within 5/32 inch of the corresponding calibration marks when the temperature is stabilized at -54°C.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

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4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

(4) Overpotential Test

(5) Dielectric Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3g at the mounting points of the case.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.1.4 Overpotential - A potential of 600 volts, rms, 60 cps applied for one second between power ground and signal ground at room temperature and approximately sea-level pressure shall result in no arcing or permanent damage to the wiring, insulation, or equipment.

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4.3.1.5 Dielectric Test - A potential of 350 volts, rms, 60 cps applied for one minute between power ground and signal ground at room temperature and approximately sea-level pressure shall result in no arcing or permanent damage to wiring, insulation, or equipment.

4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Delivered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room condictions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

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(1) Temperature - Altitude Tests

(2) Vibration Test (90 minutes cycling in each of 3 planes.)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 -- Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	26.5 ± 2.5 volts

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

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(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance shall be determined at input voltages listed in paragraph 3.3.1.5 as follows:

(a) All input voltages at nominal values

(b) All input voltages at minimum values

(c) All input voltages at maximum values

These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

6.3 Ordering Data - Procurement documents should specify the level of packaging and packing desired.

NOTICE - When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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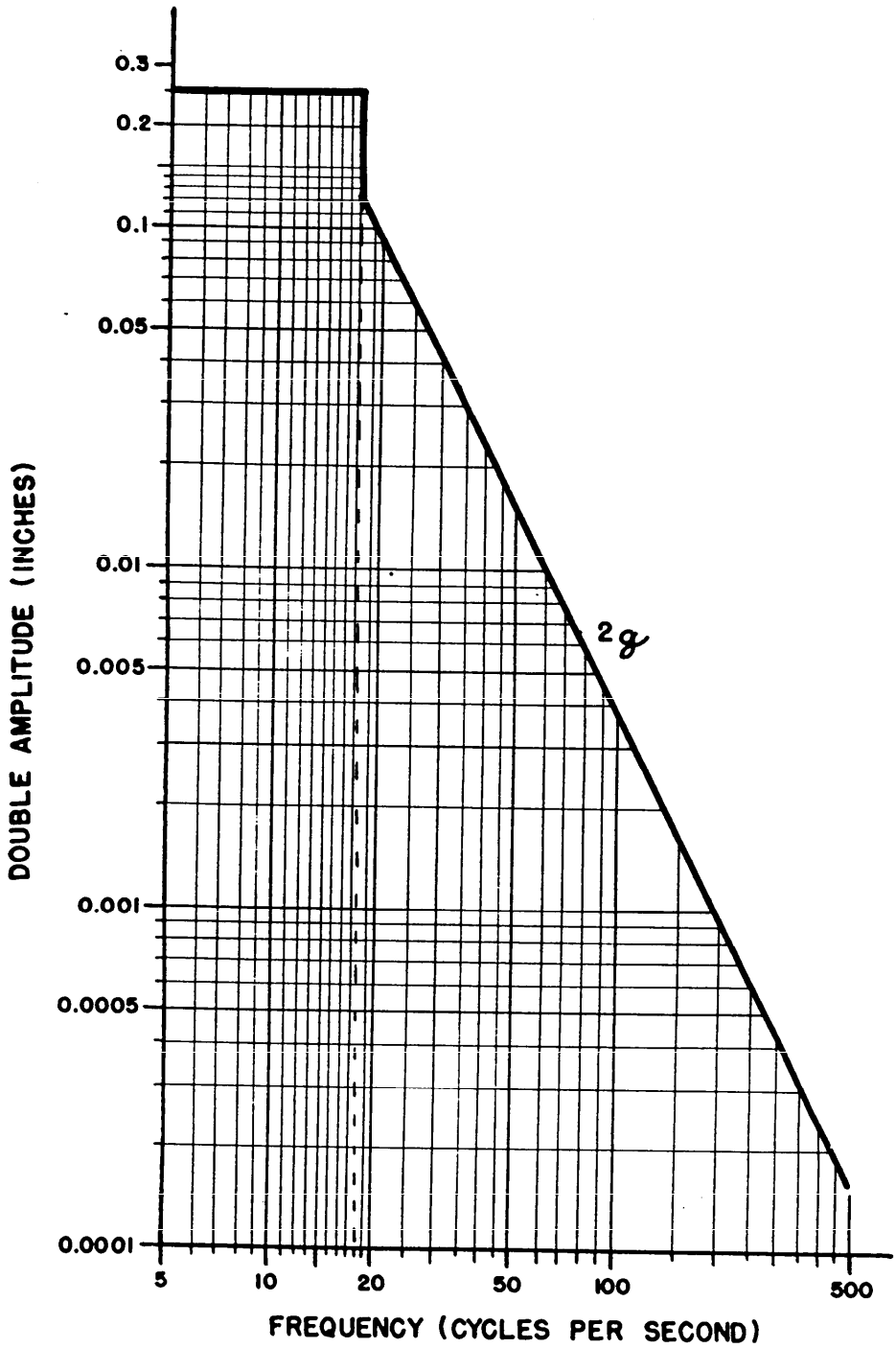


Figure 1. Vibration requirements.

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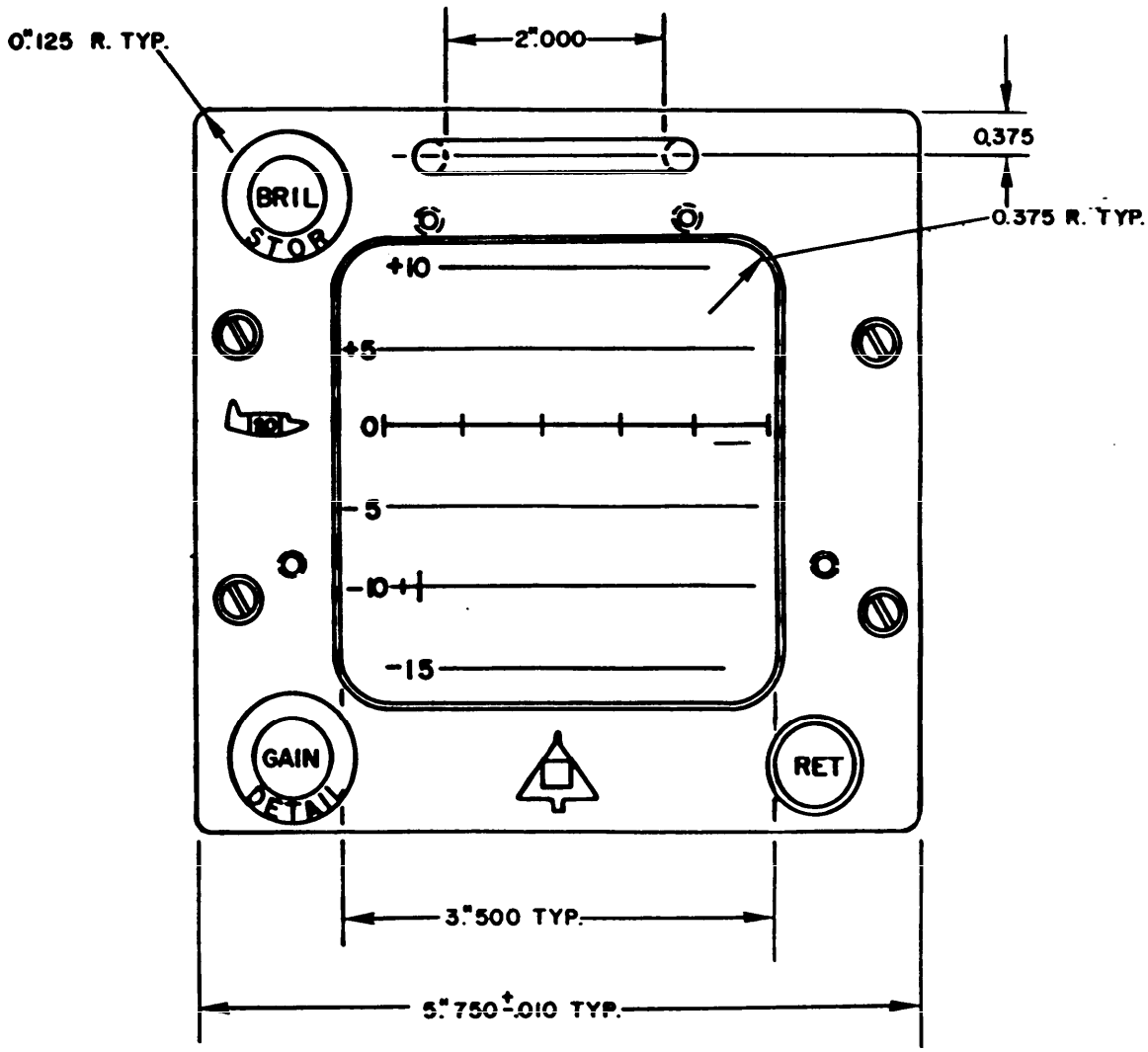


Figure 2. Front panel.

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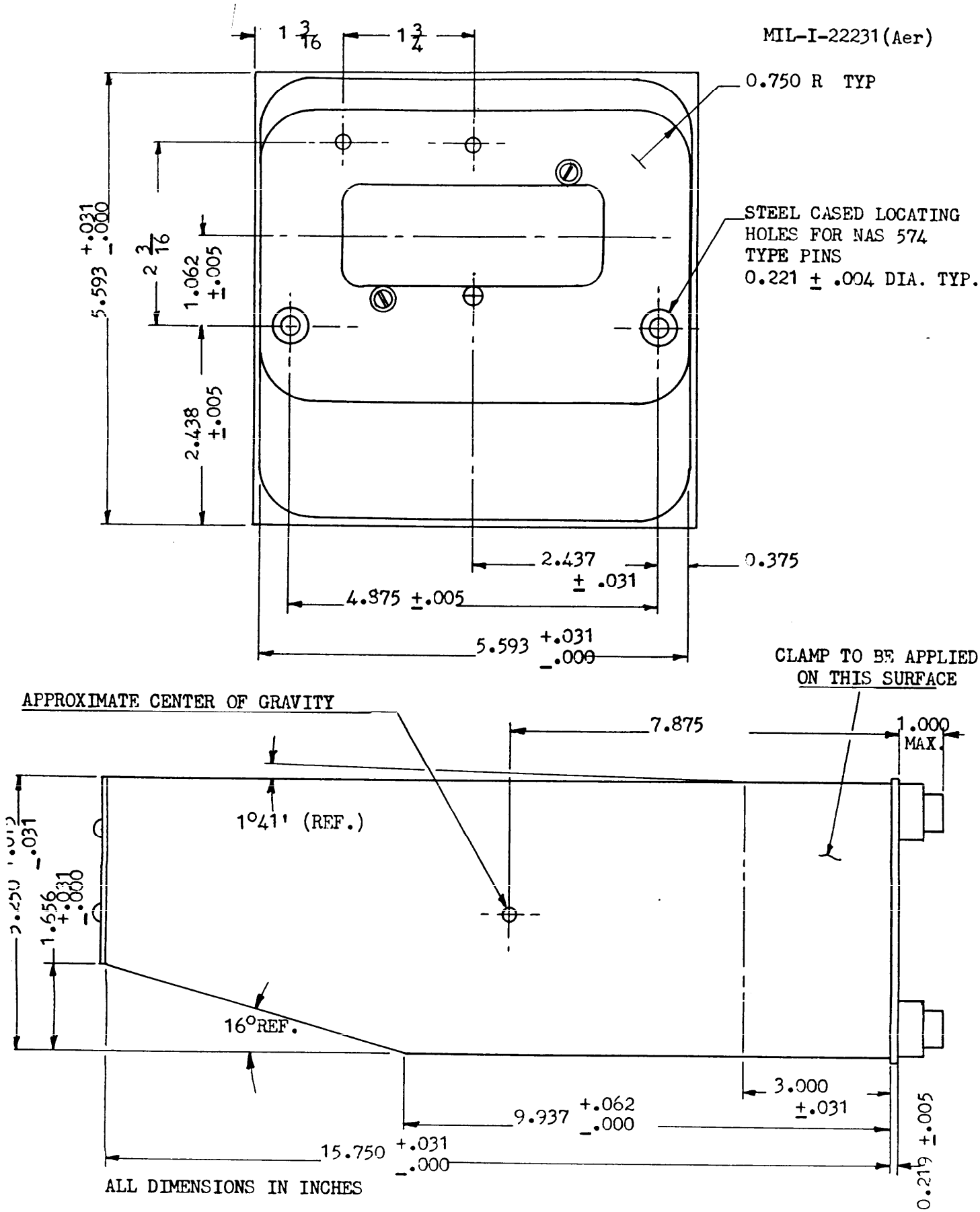


Figure 3. Case dimensions.

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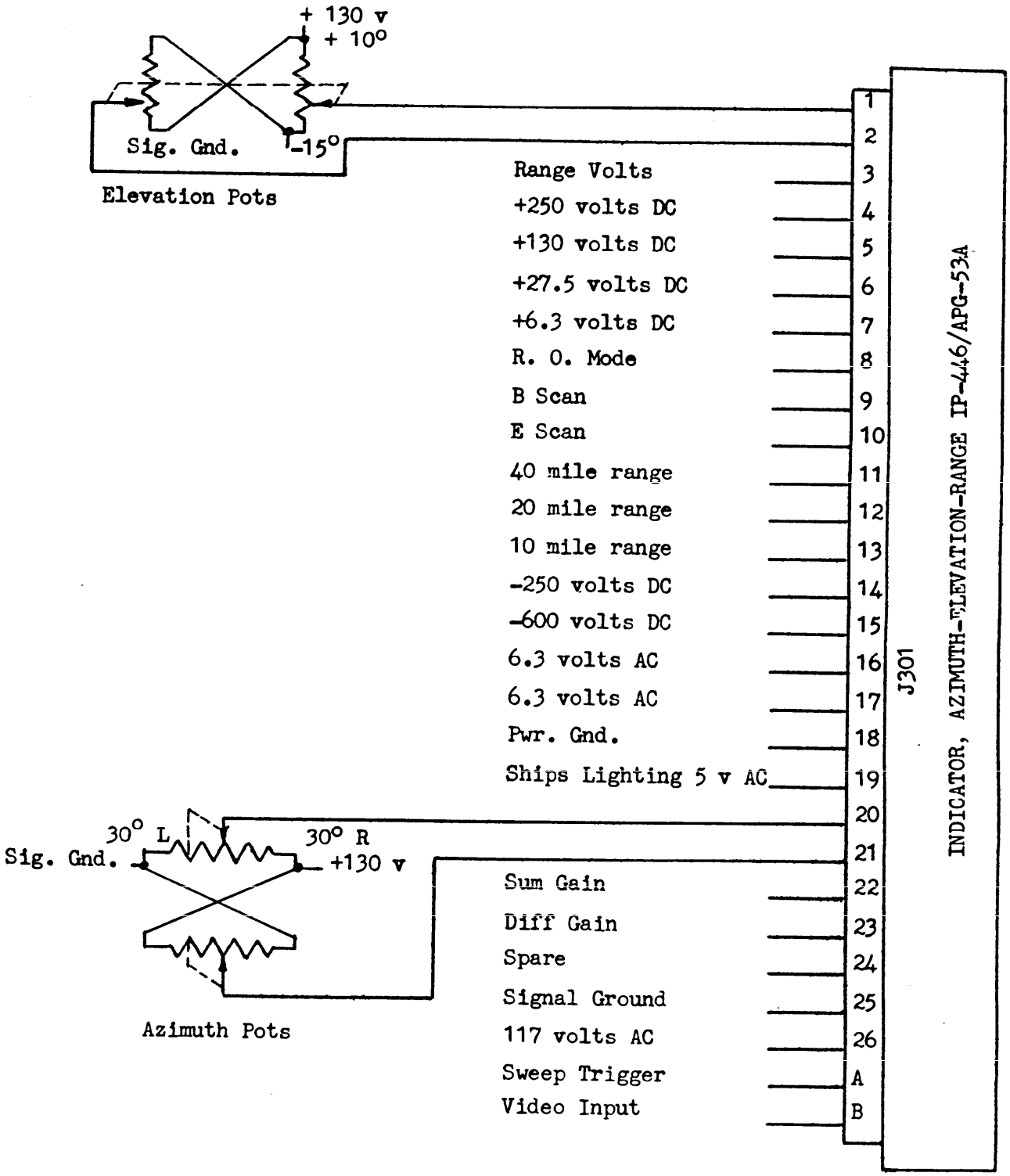


Figure 4. External connections

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